Primitive Theory Workbook: Solved and Audited Edition

Profile Sieve Empirical Likelihood with Nuisance Breaks

Complete proofs, exact rate calculations, and necessary repairs

Contents

1 Primitive Setup and Notation

For a scalar array $\boldsymbol{a} = (a_1, \ldots, a_T)'$, write

$$\|\boldsymbol{a}\|_T^2 = T^{-1} \sum_{t=1}^{T} a_t^2, \qquad \|\boldsymbol{a}\|_\infty = \max_{t \le T} |a_t|.$$

All $O_p(\cdot)$ and $o_p(\cdot)$ statements for vectors and matrices use Euclidean and spectral norm unless another norm is displayed.

The maintained null model is

$$Y_t = \beta_0 X_{t-1} + \sum_{j=0}^{q_0} m_j(W_{t-1}) \mathbf{1}\{k_{j,0} < t \le k_{j+1,0}\} + U_t, \tag{1.1}$$

where

$$0 = k_{0,0} < k_{1,0} < \cdots < k_{q_0,0} < k_{q_0+1,0} = T.$$

Throughout the primitive theory, q_0 is fixed. The true break dates are minimum-spaced: for some fixed $\epsilon > 0$,

$$\min_{0 \le j \le q_0} (k_{j+1,0} - k_{j,0}) \ge \epsilon T.$$

The coefficient β_0 is stable. All structural breaks are in the nuisance component.

For a candidate partition $\Lambda = (k_1, \ldots, k_q)$, let $k_0 = 0$, $k_{q+1} = T$, and

$$I_j(\Lambda) = \{t : k_j < t \le k_{j+1}\}, \qquad j_\Lambda(t) = j \text{ if } t \in I_j(\Lambda).$$

For the same spacing constant, define the admissible candidate set

$$\mathcal{L}_q(\epsilon) = \left\{ \Lambda = (k_1, \ldots, k_q) : 0 = k_0 < k_1 < \cdots < k_q < k_{q+1} = T,\ \min_{0 \le j \le q} (k_{j+1} - k_j) \ge \epsilon T \right\}.$$

Let

$$\mu_t = m_{j_{\Lambda_0}(t)}(W_{t-1}), \qquad \boldsymbol{\mu} = (\mu_1, \ldots, \mu_T)'.$$

Let $\boldsymbol{P}$ be the $T \times K$ sieve matrix with t-th row

$$\boldsymbol{P}_K(W_{t-1})' = (p_1(W_{t-1}), \ldots, p_K(W_{t-1})).$$

For a partition Λ, let $\boldsymbol{D}_j(\Lambda)$ be the $T \times T$ diagonal selector matrix with t-th diagonal entry $\mathbf{1}\{t \in I_j(\Lambda)\}$, $j = 0, \ldots, q$. Define

$$\boldsymbol{Q}_{K,\Lambda} = [\boldsymbol{D}_0(\Lambda)\boldsymbol{P}, \ldots, \boldsymbol{D}_q(\Lambda)\boldsymbol{P}], \qquad \boldsymbol{M}_{K,\Lambda} = \boldsymbol{I}_T - \boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}_{K,\Lambda}'\boldsymbol{Q}_{K,\Lambda})^\dagger \boldsymbol{Q}_{K,\Lambda}'. \tag{1.2}$$

The Moore–Penrose inverse makes $\boldsymbol{M}_{K,\Lambda}$ algebraically defined even when $\boldsymbol{Q}_{K,\Lambda}'\boldsymbol{Q}_{K,\Lambda}$ is singular. Such singularity can arise if a candidate segment is too short relative to K or if the selected basis columns are collinear on that segment; the projection remains defined, but the asymptotic regularity used below may fail. The theory therefore relies on minimum-spaced admissible partitions and Gram-stability/full-rank conditions on the relevant candidate sets.

For each (β, Λ), define

$$\widehat{\boldsymbol{u}}(\beta, \Lambda) = \boldsymbol{M}_{K,\Lambda}(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}), \tag{1.3}$$
$$\boldsymbol{w}^c(\Lambda) = \boldsymbol{M}_{K,\Lambda}\boldsymbol{w}, \tag{1.4}$$
$$\widehat{Z}_t(\beta, \Lambda) = \widehat{u}_t(\beta, \Lambda) w_t^c(\Lambda). \tag{1.5}$$

Let

$$S_T(\Lambda) = T^{-1/2} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda), \qquad V_T(\Lambda) = T^{-1} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda)^2.$$

For a scalar score array $Z_1, \ldots, Z_T$ satisfying the convex-hull condition, let λ_T be the unique solution of

$$\sum_{t=1}^{T} \frac{Z_t}{1 + \lambda_T Z_t} = 0, \qquad 1 + \lambda_T Z_t > 0 \quad (t \le T),$$

and define the twice negative empirical-likelihood log ratio by

$$\ell_T(Z_1, \ldots, Z_T) = 2 \sum_{t=1}^{T} \log(1 + \lambda_T Z_t). \tag{1.6}$$

In particular,

$$\ell_T(\beta, \Lambda) = \ell_T\{\widehat{Z}_1(\beta, \Lambda), \ldots, \widehat{Z}_T(\beta, \Lambda)\}.$$

For an interval $[s, e]$, define the null-imposed segment cost

$$\mathcal{C}_K(s, e; \beta) = \min_{\boldsymbol{c} \in \mathbb{R}^K} \sum_{t=s}^{e} \{Y_t - \beta X_{t-1} - \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}\}^2.$$

For $\Lambda = (k_1, \ldots, k_q)$, define

$$RSS_K(\beta, \Lambda) = \sum_{j=0}^{q} \mathcal{C}_K(k_j + 1, k_{j+1}; \beta).$$

All minimizers are measurable selections with a fixed deterministic tie-breaking rule. For known order,

$$\widehat{\Lambda}_q(\beta) = \arg \min_{\Lambda \in \mathcal{L}_q(\epsilon)} RSS_K(\beta, \Lambda), \qquad \widetilde{\Lambda} = \widehat{\Lambda}_{q_0}(\beta_0).$$

For unknown order,

$$\widehat{q}(\beta) = \arg \min_{0 \le q \le \bar{q}} \{RSS_K(\beta, \widehat{\Lambda}_q(\beta)) + \mathrm{pen}_T(q, K)\}, \qquad \widehat{\Lambda}(\beta) = \widehat{\Lambda}_{\widehat{q}(\beta)}(\beta).$$

Although the Moore–Penrose inverse makes $\boldsymbol{M}_{K,\Lambda}$ algebraically defined for arbitrary Λ, all theoretical searches are restricted to minimum-spaced admissible partitions $\mathcal{L}_q(\epsilon)$; thus no candidate segment is allowed to be asymptotically shorter than the sieve dimension K, and the Gram matrices used in the proofs are nonsingular on the relevant candidate set.

The sieve dimension is allowed to depend on the sample size and is written $K = K_T$. Throughout, $K_T \to \infty$, subject to the rate restrictions in Assumptions 3–6 and the local

Gram regularity conditions below. The basis is normalized so that its first component is constant, say $p_1(w) \equiv 1$, allowing regime-specific intercept and level shifts to be represented by the block sieve.

For local-power calculations, distinguish the null value used in the statistic from the data-generating slope. The alternative law allows

$$\beta_T = \beta_0 + d_T, \qquad d_T \to 0,$$

with data generated under P_{β_T} while the profile empirical likelihood is evaluated at β_0. Equivalently,

$$Y_t - \beta_0 X_{t-1} = \mu_t + U_t + d_T X_{t-1},$$

with the same nuisance break partition Λ_0. Section 9 records the corresponding drift conditions.

2 Primitive Assumptions

Assumption 1 (Primitive stochastic environment)**.** *The triangular array is defined on a filtered probability space* $(\Omega, \mathcal{A}, \{\mathcal{F}_t\}_{t=0}^T, \mathbb{P})$*. The variables* X_{t-1}*,* W_{t-1}*,* $\boldsymbol{P}_K(W_{t-1})$*, and* $w_t = w(X_{t-1})$ *are* $\mathcal{F}_{t-1}$*-measurable. The error satisfies*

$$\mathbb{E}(U_t \mid \mathcal{F}_{t-1}) = 0, \qquad \mathbb{E}(|U_t|^{2+\delta} \mid \mathcal{F}_{t-1}) \leq C$$

for some $\delta > 0$ *and finite* C*. The conditional variance* $\sigma_t^2 = \mathbb{E}(U_t^2 \mid \mathcal{F}_{t-1})$ *is bounded above and away from zero.*

Assumption 2 (Score admissibility under full-sample projection)**.** *For the true partition,* $\boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}$ *satisfies one of the following primitive routes.*

Route A: $\boldsymbol{v}_0$ *is admissible in the martingale CLT below.*

If the full-sample projection is not predictable, assume there is an $\mathcal{F}_{t-1}$*-measurable approximation* $v_{t,0}^*$ *such that*

$$T^{-1/2}\sum_{t=1}^{T} U_t(v_{t,0} - v_{t,0}^*) = o_p(1), \qquad T^{-1}\sum_{t=1}^{T} U_t^2\{v_{t,0}^2 - (v_{t,0}^*)^2\} = o_p(1).$$

Assumption 3 (Bounded weights, nuisance functions, and basis envelope)**.** *The sieve dimension satisfies* $K = K_T \to \infty$*. The first basis element is constant,* $p_1(w) \equiv 1$*. There are constants* $C_w, C_m < \infty$ *and an envelope* ζ_K *such that*

$$\max_{t \leq T} |w_t| \leq C_w, \qquad \max_{0 \leq j \leq q_0} \sup_x |m_j(x)| \leq C_m, \qquad \max_{t \leq T} \|\boldsymbol{P}_K(W_{t-1})\| \leq \zeta_K$$

with probability tending to one.

Assumption 4 (Spacing, smoothness, and break detectability)**.** *The number q_0 is fixed. For some $\epsilon > 0$,*

$$\min_{0\le j\le q_0}(k_{j+1,0} - k_{j,0}) \ge \epsilon T.$$

Each m_j belongs to a common smoothness class $\mathcal{H}^s$. If $q_0 \ge 1$, then

$$\Delta_T = \min_{0\le j<q_0} \|m_{j+1} - m_j\|_{L^2(P_W)} > 0, \qquad \frac{T\Delta_T^2}{K\log T} \to \infty.$$

Assumption 5 (Uniform Gram stability)**.** *For the true partition and all candidate partitions in the local neighborhood*

$$\mathcal{N}_T(r_T) = \{\Lambda : \max_j |k_j - k_{j,0}| \le Cr_T,\ q = q_0\},$$

the eigenvalues of $T^{-1}\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda}$, restricted to nonzero directions, are bounded away from zero and infinity with probability tending to one.

Assumption 6 (Sieve approximation and projection bias)**.** *For each regime j, there is a coefficient vector $\boldsymbol{c}_{j,K}$ such that*

$$m_j(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_{j,K} + r_{j,K,t}, \qquad \max_j \|\boldsymbol{r}_{j,K}\|_T = O_p(a_K), \qquad \max_j \|\boldsymbol{r}_{j,K}\|_\infty = O_p(a_K).$$

Let $\boldsymbol{r}_0$ be the stacked true-partition approximation error. The rates satisfy

$$T^{-1/2}\boldsymbol{r}_0'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} = o_p(1), \qquad a_K(1+\zeta_K) \to 0, \qquad a_K(1+\zeta_K)^2 = o(\sqrt{T}).$$

Also,

$$\frac{(q_0+1)K}{\sqrt{T}} \to 0, \qquad \zeta_K\sqrt{\frac{(q_0+1)K}{T}} \to 0.$$

Assumption 7 (Fixed-order localization target)**.** *For known q_0,*

$$\max_{1\le j\le q_0} |\widetilde{k}_j - k_{j,0}| = O_p(r_T), \qquad r_T = o(\sqrt{T}).$$

This assumption is a target to be derived in Section 7; it may be used as an input in Section 6.

Assumption 8 (Empirical-likelihood convex hull)**.** *For the score array under consideration,*

$$\mathbb{P}\{\min_{t\le T} Z_{t,T} < 0 < \max_{t\le T} Z_{t,T}\} \to 1.$$

When used for the true partition, $Z_{t,T} = \widehat{Z}_t(\beta_0, \Lambda_0)$. When used for the estimated partition, $Z_{t,T} = \widehat{Z}_t(\beta_0, \widetilde{\Lambda})$.

3 Rigour Audit and Completing Conditions

Several displays in the original workbook were labelled "targets." They are not all consequences of Assumptions 1–8 as originally written. The precise gaps are: the predictable approximation needs an envelope and variance-replacement condition; local partition perturbations need local stability of the sieve bias; logarithmic RSS concentration does not follow from only a finite $(2+\delta)$-moment; and an overfitting penalty that merely diverges need not dominate the searched RSS gain. The following conditions make every proof below logically complete. They are stated separately so that none of the missing hypotheses is hidden.

Write

$$d_K = (q_0+1)K, \qquad b_K = 1+\zeta_K, \qquad \delta_0 = \min\{\delta, 2\}.$$

Assumption 9 (Oracle envelope completion)**.** *When the predictable-approximation route in Assumption 2 is used, the approximation can be chosen so that*

$$\max_{t\le T} |v^*_{t,0}| = O_p(b_K), \qquad T^{-1}\sum_{t=1}^{T} \sigma_t^2\{(v^*_{t,0})^2 - v^2_{t,0}\} = o_p(1).$$

Moreover,

$$b_K^{2+\delta_0} T^{-\delta_0/2} \longrightarrow 0. \tag{3.1}$$

If $\boldsymbol{v}_0$ itself is predictable, take $v^*_{t,0} = v_{t,0}$.

Assumption 10 (Local sieve-bias stability)**.** *In the one-break section, for every fixed $C<\infty$,*

$$\sup_{|k-k_0|\le Cr_T} T^{-1/2}\, |\boldsymbol{r}_0'\{\boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}\}\boldsymbol{w}| = o_p(1). \tag{3.2}$$

A sufficient primitive rate, proved in the solution of Lemma 9, is

$$\frac{a_K b_K (1+\zeta_K^2) r_T}{\sqrt{T}} \longrightarrow 0. \tag{3.3}$$

Assumption 11 (Local full-rank series regularity)**.** *In the one-break section, every segment Gram matrix associated with a partition satisfying $|k-k_0|\le Cr_T$ is nonsingular, and its eigenvalues are between cT and CT, uniformly with probability tending to one. Also*

$$\zeta_K^2 = O(K). \tag{3.4}$$

The last display is the usual normalized-series envelope condition. It is needed for the denominator rate stated in Lemma 11; without it that rate must be replaced by the larger order $b_K^2\zeta_K^2 r_T/T$.

Assumption 12 (Localization concentration completion)**.** *This assumption is used only in Section 7. Given the design sigma-field $\mathcal{D}_T$, the errors are independent, centered, conditionally sub-Gaussian with a common conditional variance σ^2. Every minimum-spaced one-break* *sieve matrix has rank $2K$. If $h = |k - k_0|$ and $\boldsymbol{A}_k = \boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}$, then, uniformly over candidates,*

$$\boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{\mu} \geq c_\Delta h\Delta_T^2 - \rho_T(k), \qquad \sup_{k:\, h\geq 1} \frac{|\rho_T(k)|}{h\Delta_T^2} = o_p(1), \tag{3.5}$$

$$\|\boldsymbol{A}_k\boldsymbol{\mu}\|^2 \leq ChK \tag{3.6}$$

with probability tending to one. Condition (3.5) is a sample profile-identification condition. It follows, for example, in the exact-sieve case from uniform segment Gram bounds and a local empirical jump-energy lower bound; this is verified explicitly in the proof of Lemma 12.

Remark 1 (Why the localization completion is necessary)**.** The martingale restriction $\mathbb{E}(U_t \mid \mathcal{F}_{t-1}) = 0$ does not imply $\mathbb{E}(U_t \mid \mathcal{D}_T) = 0$, because $\mathcal{D}_T$ contains future regressors and those regressors may reveal U_t. Also, a finite $(2+\delta)$-moment does not by itself yield a $\log T$ search bound for a quadratic form. Thus Lemmas 12 and 13 genuinely require an additional route such as Assumption 12.

Lemma 1 (Scalar empirical-likelihood expansion)**.** *Let $Z_{1,T}, \ldots, Z_{T,T}$ satisfy the convex-hull condition and suppose*

$$A_T = T^{-1/2}\sum_{t=1}^{T} Z_{t,T} = O_p(1), \qquad B_T = T^{-1}\sum_{t=1}^{T} Z_{t,T}^2 \geq c + o_p(1)$$

for some $c > 0$, and

$$\max_{t\leq T} |Z_{t,T}| = o_p(\sqrt{T}).$$

Then

$$\ell_T(Z_{1,T}, \ldots, Z_{T,T}) = \frac{A_T^2}{B_T} + o_p(1). \tag{3.7}$$

Proof. Put $S_1 = \sum_t Z_{t,T}$, $S_2 = \sum_t Z_{t,T}^2$, and $M_T = \max_t |Z_{t,T}|$. The convex-hull condition gives a unique multiplier λ_T in the interval on which every $1 + \lambda_T Z_{t,T}$ is positive. Its score equation implies the exact identity

$$S_1 = \lambda_T \sum_{t=1}^{T} \frac{Z_{t,T}^2}{1 + \lambda_T Z_{t,T}}.$$

Consequently,

$$|S_1| \geq \frac{|\lambda_T| S_2}{1 + |\lambda_T| M_T}.$$

After rearrangement,

$$|\lambda_T|\{S_2 - |S_1| M_T\} \leq |S_1|.$$

Now $|S_1| = O_p(\sqrt{T})$, $S_2 \geq (c + o_p(1))T$, and $|S_1| M_T = o_p(T)$. Hence

$$\lambda_T = O_p(T^{-1/2}), \qquad |\lambda_T| M_T = o_p(1).$$

Using $(1+x)^{-1} = 1 - x + x^2/(1+x)$ in the score equation gives

$$0 = S_1 - \lambda_T S_2 + R_{1T}, \qquad |R_{1T}| \le \frac{\lambda_T^2 M_T S_2}{1 - |\lambda_T| M_T}.$$

Thus

$$\lambda_T = \frac{S_1}{S_2} + o_p(T^{-1/2}).$$

Finally, Taylor expansion of the logarithm, uniformly because $|\lambda_T| M_T = o_p(1)$, yields

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = 2\lambda_T S_1 - \lambda_T^2 S_2 + R_{2T},$$

where

$$|R_{2T}| \le C |\lambda_T|^3 M_T S_2 = o_p(1).$$

Substituting the expansion for λ_T gives

$$2 \sum_t \log(1 + \lambda_T Z_{t,T}) = \frac{S_1^2}{S_2} + o_p(1) = \frac{A_T^2}{B_T} + o_p(1),$$

which proves (3.7). $\square$

4 Deterministic Projection Algebra

Lemma 2 (Block projection identities)**.** *For every partition* Λ*,*

$$\boldsymbol{Q}'_{K,\Lambda} \boldsymbol{M}_{K,\Lambda} = \boldsymbol{0}, \qquad \boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}, \qquad \boldsymbol{M}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}.$$

Consequently, for every conformable $\boldsymbol{c}$*,*

$$(\boldsymbol{Q}_{K,\Lambda} \boldsymbol{c})' \boldsymbol{M}_{K,\Lambda} \boldsymbol{w} = 0.$$

Detailed solution. Abbreviate $\boldsymbol{Q} = \boldsymbol{Q}_{K,\Lambda}$. Take a thin singular-value decomposition

$$\boldsymbol{Q} = \boldsymbol{U}_r \boldsymbol{\Sigma}_r \boldsymbol{V}'_r,$$

where $r = \operatorname{rank}(\boldsymbol{Q})$, the columns of $\boldsymbol{U}_r$ and $\boldsymbol{V}_r$ are orthonormal, and $\boldsymbol{\Sigma}_r$ is diagonal with strictly positive entries. Then

$$\boldsymbol{Q}'\boldsymbol{Q} = \boldsymbol{V}_r \boldsymbol{\Sigma}_r^2 \boldsymbol{V}'_r, \qquad (\boldsymbol{Q}'\boldsymbol{Q})^\dagger = \boldsymbol{V}_r \boldsymbol{\Sigma}_r^{-2} \boldsymbol{V}'_r.$$

Therefore

$$\boldsymbol{\Pi}_{K,\Lambda} = \boldsymbol{Q}(\boldsymbol{Q}'\boldsymbol{Q})^\dagger \boldsymbol{Q}' = \boldsymbol{U}_r \boldsymbol{U}'_r.$$

This is the orthogonal projector onto $\operatorname{col}(\boldsymbol{Q})$. In particular,

$$\boldsymbol{\Pi}'_{K,\Lambda} = \boldsymbol{\Pi}_{K,\Lambda}, \qquad \boldsymbol{\Pi}^2_{K,\Lambda} = \boldsymbol{\Pi}_{K,\Lambda}, \qquad \boldsymbol{\Pi}_{K,\Lambda} \boldsymbol{Q} = \boldsymbol{Q}.$$

Since $\boldsymbol{M}_{K,\Lambda} = \boldsymbol{I}_T - \boldsymbol{\Pi}_{K,\Lambda}$,

$$\boldsymbol{M}'_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}$$

and

$$\boldsymbol{M}^2_{K,\Lambda} = (\boldsymbol{I}_T - \boldsymbol{\Pi}_{K,\Lambda})^2 = \boldsymbol{I}_T - 2\boldsymbol{\Pi}_{K,\Lambda} + \boldsymbol{\Pi}^2_{K,\Lambda} = \boldsymbol{M}_{K,\Lambda}.$$

Also

$$\boldsymbol{Q}'\boldsymbol{M}_{K,\Lambda} = \{\boldsymbol{M}_{K,\Lambda}\boldsymbol{Q}\}' = \boldsymbol{0}'.$$

For conformable $\boldsymbol{c}$ and $\boldsymbol{w}$, it follows immediately that

$$(\boldsymbol{Q}\boldsymbol{c})'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = \boldsymbol{c}'\boldsymbol{Q}'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w} = 0.$$

□

Lemma 3 (Uniform fitted-value and leverage bounds)**.** *Under Assumptions 3 and 5, uniformly over $\Lambda \in \mathcal{N}_T(r_T)$,*

$$\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\|_\infty = O_p(1+\zeta_K), \qquad \max_{t\le T}\{\boldsymbol{Q}_{K,\Lambda}(\boldsymbol{Q}'_{K,\Lambda}\boldsymbol{Q}_{K,\Lambda})^\dagger\boldsymbol{Q}'_{K,\Lambda}\}_{tt} = O_p(\zeta_K^2 K/T).$$

Detailed solution. Write $\boldsymbol{Q} = \boldsymbol{Q}_{K,\Lambda}$, $\boldsymbol{G} = \boldsymbol{Q}'\boldsymbol{Q}$, and let $\boldsymbol{q}'_t$ denote row t of $\boldsymbol{Q}$. Because exactly one regime block is active in each row,

$$\|\boldsymbol{q}_t\| = \|\boldsymbol{P}_K(W_{t-1})\| \le \zeta_K.$$

On the uniform Gram event in Assumption 5,

$$\|\boldsymbol{G}^\dagger\| \le \frac{C}{T}, \qquad \|\boldsymbol{Q}\|^2 = \lambda_{\max}(\boldsymbol{Q}'\boldsymbol{Q}) \le CT.$$

Since $\|\boldsymbol{w}\| \le C_w\sqrt{T}$, the least-squares coefficient vector

$$\widehat{\boldsymbol{c}}_\Lambda = \boldsymbol{G}^\dagger\boldsymbol{Q}'\boldsymbol{w}$$

satisfies

$$\|\widehat{\boldsymbol{c}}_\Lambda\| \le \frac{C}{T}\,\|\boldsymbol{Q}\|\,\|\boldsymbol{w}\| \le C.$$

Thus, uniformly in t and Λ,

$$|(\boldsymbol{\Pi}_{K,\Lambda}\boldsymbol{w})_t| = |\boldsymbol{q}'_t\widehat{\boldsymbol{c}}_\Lambda| \le C\zeta_K,$$

and hence

$$\|\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}\|_\infty \le \|\boldsymbol{w}\|_\infty + \|\boldsymbol{\Pi}_{K,\Lambda}\boldsymbol{w}\|_\infty = O_p(1+\zeta_K).$$

For the leverage,

$$(\boldsymbol{\Pi}_{K,\Lambda})_{tt} = \boldsymbol{q}'_t\boldsymbol{G}^\dagger\boldsymbol{q}_t \le \|\boldsymbol{G}^\dagger\|\,\|\boldsymbol{q}_t\|^2 \le C\frac{\zeta_K^2}{T}.$$

This is sharper than the displayed target and therefore implies $O_p(\zeta_K^2K/T)$, because $K \ge 1$. All inequalities hold on a single uniform Gram event whose probability tends to one. □

5 Known-Partition Primitive Wilks

Set

$$\boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}, \qquad A_T^{(q)} = T^{-1/2}\sum_{t=1}^{T} U_t v_{t,0}, \qquad V_T^{(q)} = T^{-1}\sum_{t=1}^{T} U_t^2 v_{t,0}^2.$$

Lemma 4 (Oracle martingale CLT target)**.** *Under Assumptions 1, 2, 3, 5, and 9, suppose*

$$T^{-1}\sum_{t=1}^{T}\sigma_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \qquad \mathbb{P}(\eta_q \geq \sqrt{c_\eta}) = 1$$

for some $c_\eta > 0$. Then

$$A_T^{(q)} \xrightarrow{\text{st}} \eta_q Z,$$

where $Z \sim N(0,1)$ is independent of the limiting sigma-field generating η_q.

Detailed solution. Let $\delta_0 = \min\{\delta, 2\}$. First use the predictable route. Put

$$X_{t,T} = T^{-1/2}U_t v_{t,0}^*.$$

Then $X_{t,T}$ is a martingale-difference array. By Assumption 9 and the variance-replacement condition,

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}(X_{t,T}^2 \mid \mathcal{F}_{t-1}) &= T^{-1}\sum_{t=1}^{T}\sigma_t^2 (v_{t,0}^*)^2 \\ &= T^{-1}\sum_{t=1}^{T}\sigma_t^2 v_{t,0}^2 + o_p(1) \xrightarrow{p} \eta_q^2.\end{aligned}$$

For the conditional Lyapunov term, the conditional moment bound gives

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}(|X_{t,T}|^{2+\delta_0} \mid \mathcal{F}_{t-1}) &\leq CT^{-(1+\delta_0/2)}\sum_{t=1}^{T}|v_{t,0}^*|^{2+\delta_0} \\ &\leq Cb_K^{2+\delta_0}T^{-\delta_0/2} = o_p(1).\end{aligned}$$

The martingale triangular-array stable CLT therefore yields

$$T^{-1/2}\sum_{t=1}^{T} U_t v_{t,0}^* \xrightarrow{\text{st}} \eta_q Z,$$

where Z is standard normal and independent of the limiting sigma-field that contains η_q. Assumption 2 gives

$$T^{-1/2}\sum_{t=1}^{T} U_t(v_{t,0} - v_{t,0}^*) = o_p(1).$$

Stable Slutsky therefore gives the asserted limit for $A_T^{(q)}$. Under Route A the same argument is exactly the admissible martingale CLT postulated there, so no replacement is required. $\square$

Lemma 5 (Oracle denominator LLN target)**.** *Under the primitive stochastic and score-admissibility assumptions and Assumption 9,*

$$V_T^{(q)} - \eta_q^2 = o_p(1).$$

Detailed solution. Again use the predictable approximation; the direct predictable case is the special case $v_{t,0}^* = v_{t,0}$. Decompose

$$\begin{aligned} T^{-1}\sum_{t=1}^{T} U_t^2 (v_{t,0}^*)^2 = T^{-1}\sum_{t=1}^{T} \sigma_t^2 (v_{t,0}^*)^2 \\ + T^{-1}\sum_{t=1}^{T} (U_t^2 - \sigma_t^2)(v_{t,0}^*)^2. \end{aligned}$$

The first term converges in probability to η_q^2 by Assumption 9 and the variance convergence in Lemma 4.

For the second term, set

$$D_{t,T} = (U_t^2 - \sigma_t^2)(v_{t,0}^*)^2, \qquad p = 1 + \delta_0/2 \in (1, 2].$$

Then $D_{t,T}$ is a martingale difference and, using $2p = 2 + \delta_0$,

$$\mathbb{E}(|D_{t,T}|^p \mid \mathcal{F}_{t-1}) \le C |v_{t,0}^*|^{2p}.$$

The martingale von Bahr–Esseen inequality gives

$$\begin{aligned} \mathbb{E}\left| T^{-1}\sum_{t=1}^{T} D_{t,T} \right|^p &\le C T^{-p} \sum_{t=1}^{T} \mathbb{E}|D_{t,T}|^p \\ &\le C b_K^{2+\delta_0} T^{-\delta_0/2} = o(1), \end{aligned}$$

where the usual localization to the event $\max_t |v_{t,0}^*| \le C b_K$ is understood. Hence the martingale term is $o_p(1)$. Finally, Assumption 2 supplies

$$T^{-1}\sum_{t=1}^{T} U_t^2 \{ v_{t,0}^2 - (v_{t,0}^*)^2 \} = o_p(1),$$

so replacing $v_{t,0}^*$ by $v_{t,0}$ proves $V_T^{(q)} = \eta_q^2 + o_p(1)$. □

Lemma 6 (Oracle maximum bound target)**.** *Under the primitive moment and leverage conditions and Assumption 9,*

$$\max_{t \le T} |U_t v_{t,0}| = o_p(\sqrt{T}).$$

Detailed solution. Lemma 3 gives

$$\max_{t \le T} |v_{t,0}| = O_p(b_K).$$

Fix $C < \infty$ and work on the event $\max_t |v_{t,0}| \le Cb_K$. For every $\varepsilon > 0$, the union bound and the $(2+\delta_0)$-moment bound imply

$$
\begin{aligned}
\mathbb{P}\left(\max_{t\le T} |U_t v_{t,0}| > \varepsilon\sqrt{T}\right) &\le \sum_{t=1}^{T} \mathbb{P}\left(|U_t| > \frac{\varepsilon\sqrt{T}}{Cb_K}\right) + o(1) \\
&\le C_\varepsilon T \left(\frac{b_K}{\sqrt{T}}\right)^{2+\delta_0} + o(1) \\
&= C_\varepsilon b_K^{2+\delta_0} T^{-\delta_0/2} + o(1) \longrightarrow 0.
\end{aligned}
$$

Thus $\max_t |U_t v_{t,0}|/\sqrt{T} \to_p 0$. Equivalently, a sufficient rate is

$$
b_K T^{1/(2+\delta_0)-1/2} \to 0,
$$

which is exactly (3.1) written after taking a $(2+\delta_0)$-th root. □

Lemma 7 (Feasible true-partition replacement)**.** *Under Assumptions 1, 6, 3, 5, and 9,*

$$
T^{-1/2} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda_0) = T^{-1/2} \sum_{t=1}^{T} U_t v_{t,0} + o_p(1), \tag{5.1}
$$

$$
T^{-1} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda_0)^2 = T^{-1} \sum_{t=1}^{T} U_t^2 v_{t,0}^2 + o_p(1), \tag{5.2}
$$

$$
\max_{t\le T} |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}). \tag{5.3}
$$

Detailed solution. Let $\boldsymbol{Q}_0 = \boldsymbol{Q}_{K,\Lambda_0}$, $\boldsymbol{\Pi}_0 = \boldsymbol{I}_T - \boldsymbol{M}_{K,\Lambda_0}$, and $\boldsymbol{v}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}$. The sieve representation gives

$$
\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{Q}_0 \boldsymbol{c}_0 + \boldsymbol{r}_0 + \boldsymbol{U}.
$$

By Lemma 2,

$$
\widehat{\boldsymbol{u}}(\beta_0, \Lambda_0) = \boldsymbol{M}_{K,\Lambda_0}(\boldsymbol{r}_0 + \boldsymbol{U}).
$$

Define

$$
\boldsymbol{e}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{r}_0, \qquad \boldsymbol{h}_0 = \boldsymbol{\Pi}_0 \boldsymbol{U}.
$$

Then, componentwise,

$$
\widehat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + (e_{t,0} - h_{t,0}) v_{t,0}. \tag{5.4}
$$

We first establish the projection-noise bounds rather than assuming them. Because each row of $\boldsymbol{Q}_0$ is predictable, martingale orthogonality and the Gram upper bound imply

$$
\|\boldsymbol{Q}_0' \boldsymbol{U}\| = O_p(\sqrt{T d_K}).
$$

Indeed, after localization to the Gram event,

$$
\mathbb{E}\|\boldsymbol{Q}_0' \boldsymbol{U}\|^2 = \sum_{t=1}^{T} \mathbb{E}\{\sigma_t^2 \|\boldsymbol{q}_{t,0}\|^2\} \le C\mathbb{E}\,\mathrm{tr}(\boldsymbol{Q}_0' \boldsymbol{Q}_0) = O(T d_K).
$$

Since $\|(\boldsymbol{Q}_0'\boldsymbol{Q}_0)^\dagger\| = O_p(T^{-1})$,

$$\begin{aligned}\|\boldsymbol{h}_0\|_T^2 &= T^{-1}\boldsymbol{U}'\boldsymbol{\Pi}_0\boldsymbol{U}\\ &\le \frac{C}{T^2}\|\boldsymbol{Q}_0'\boldsymbol{U}\|^2 = O_p(d_K/T),\end{aligned}$$

and, row by row,

$$\|\boldsymbol{h}_0\|_\infty \le \zeta_K\|(\boldsymbol{Q}_0'\boldsymbol{Q}_0)^\dagger\|\|\boldsymbol{Q}_0'\boldsymbol{U}\| = O_p\left(\zeta_K\sqrt{d_K/T}\right).$$

For the approximation term, orthogonal projection is a contraction, so

$$\|\boldsymbol{e}_0\|_T \le \|\boldsymbol{r}_0\|_T = O_p(a_K).$$

Furthermore,

$$\|(\boldsymbol{Q}_0'\boldsymbol{Q}_0)^\dagger\boldsymbol{Q}_0'\boldsymbol{r}_0\| \le \frac{C}{T}\|\boldsymbol{Q}_0\|\,\|\boldsymbol{r}_0\| = O_p(a_K),$$

which gives

$$\|\boldsymbol{e}_0\|_\infty \le \|\boldsymbol{r}_0\|_\infty + \|\boldsymbol{\Pi}_0\boldsymbol{r}_0\|_\infty = O_p\{a_K(1+\zeta_K)\}.$$

Combining these bounds with $\|\boldsymbol{v}_0\|_\infty = O_p(b_K)$,

$$\begin{aligned}\|(\boldsymbol{e}_0-\boldsymbol{h}_0)\odot\boldsymbol{v}_0\|_T &\le O_p\left(a_Kb_K + b_K\sqrt{d_K/T}\right) = o_p(1),\\ \|(\boldsymbol{e}_0-\boldsymbol{h}_0)\odot\boldsymbol{v}_0\|_\infty &\le O_p\left(a_Kb_K^2 + \zeta_Kb_K\sqrt{d_K/T}\right) = o_p(\sqrt{T}).\end{aligned}$$

The first conclusion uses Assumption 6; the second uses both Assumptions 6 and 9.

For the numerator, symmetry and idempotence give an exact cancellation:

$$\begin{aligned}T^{-1/2}\sum_t \widehat{Z}_t(\beta_0,\Lambda_0) &= T^{-1/2}(\boldsymbol{U}+\boldsymbol{r}_0)'\boldsymbol{M}_{K,\Lambda_0}^2\boldsymbol{w}\\ &= T^{-1/2}\boldsymbol{U}'\boldsymbol{v}_0 + T^{-1/2}\boldsymbol{r}_0'\boldsymbol{v}_0\\ &= T^{-1/2}\boldsymbol{U}'\boldsymbol{v}_0 + o_p(1).\end{aligned}$$

For the denominator, let $\boldsymbol{d}_0 = (\boldsymbol{e}_0-\boldsymbol{h}_0)\odot\boldsymbol{v}_0$. Then

$$\begin{aligned}\left|T^{-1}\sum_t \widehat{Z}_t^2 - T^{-1}\sum_t U_t^2v_{t,0}^2\right| &\le 2\|\boldsymbol{U}\odot\boldsymbol{v}_0\|_T\|\boldsymbol{d}_0\|_T + \|\boldsymbol{d}_0\|_T^2\\ &= o_p(1),\end{aligned}$$

because Lemma 5 makes the first norm $O_p(1)$. Finally, (5.4), Lemma 6, and $\|\boldsymbol{d}_0\|_\infty = o_p(\sqrt{T})$ prove the maximum assertion. □

Theorem 1 (Primitive known-partition Wilks target)**.** *Under the primitive assumptions in this section, Assumption 9, the stated variance convergence, and the convex-hull condition for $\widehat{Z}_t(\beta_0,\Lambda_0)$,*

$$\ell_T(\beta_0,\Lambda_0) \Rightarrow \chi_1^2.$$

Detailed solution. Set

$$A_T = T^{-1/2} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda_0), \qquad B_T = T^{-1} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0, \Lambda_0)^2.$$

Lemma 7, followed by Lemmas 4 and 5, gives the joint limits

$$A_T \xrightarrow{\text{st}} \eta_q Z, \qquad B_T \xrightarrow{p} \eta_q^2.$$

Because $\eta_q^2 \geq c_\eta$ almost surely, B_T is bounded away from zero with probability tending to one. Lemmas 6 and 7 give

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0)| = o_p(\sqrt{T}).$$

The convex-hull assumption permits application of Lemma 1; hence

$$\ell_T(\beta_0, \Lambda_0) = \frac{A_T^2}{B_T} + o_p(1).$$

Stable Slutsky and positivity of η_q imply

$$\frac{A_T^2}{B_T} \Rightarrow \frac{\eta_q^2 Z^2}{\eta_q^2} = Z^2 \sim \chi_1^2.$$

This proves the theorem. □

6 One-Break Estimated-Partition Bridge

In this section only, take $q_0 = 1$, $\Lambda_0 = (k_0)$, and $\widetilde{\Lambda} = (\widetilde{k})$. Let

$$\Delta_M = \boldsymbol{M}_{K,\widetilde{\Lambda}} - \boldsymbol{M}_{K,\Lambda_0}, \qquad \mathcal{M}_T = \{t : j_{\widetilde{\Lambda}}(t) \neq j_{\Lambda_0}(t)\}.$$

Lemma 8 (Boundary-set size)**.** *If $|\widetilde{k} - k_0| = O_p(r_T)$, then*

$$|\mathcal{M}_T| = O_p(r_T).$$

Detailed solution. Suppose first that $\widetilde{k} \geq k_0$. For $t \leq k_0$, both partitions assign observation t to the left regime; for $t > \widetilde{k}$, both assign it to the right regime. The assignments differ exactly for

$$k_0 < t \leq \widetilde{k}.$$

Thus $|\mathcal{M}_T| = \widetilde{k} - k_0$. If $\widetilde{k} < k_0$, the differing observations are exactly $\widetilde{k} < t \leq k_0$, and again $|\mathcal{M}_T| = |\widetilde{k} - k_0|$. Therefore

$$|\mathcal{M}_T| = |\widetilde{k} - k_0| = O_p(r_T).$$

□

Lemma 9 (Nuisance part of partition perturbation)**.** *Under bounded nuisance functions, bounded weights, uniform Gram stability, Assumption 10, and* $|\widetilde{k}-k_0| = O_p(r_T)$*,*

$$T^{-1/2}\boldsymbol{\mu}'\Delta_M \boldsymbol{w} = O_p\biggl(\frac{(1+\zeta_K)r_T}{\sqrt{T}}\biggr) + o_p(1).$$

In particular, this term is $o_p(1)$ *if* $(1+\zeta_K)r_T/\sqrt{T} \to 0$*.*

Detailed solution. The original primitive assumptions alone do not control the term generated by moving the sieve approximation error from one projection to another; Assumption 10 supplies exactly that missing control. We now prove the stated rate.

Work first on the event $\widetilde{k} = k_0 + h$ with $h \geq 0$, and let $H = \{k_0+1, \dots, k_0+h\}$. Let $\boldsymbol{c}_0$ and $\boldsymbol{c}_1$ be the two true-regime sieve coefficients. Form a coefficient vector for the candidate partition by assigning $\boldsymbol{c}_0$ to its left block and $\boldsymbol{c}_1$ to its right block. Then

$$\boldsymbol{\mu} = \boldsymbol{Q}_{K,\widetilde{\Lambda}}\boldsymbol{c}^{(\widetilde{k})} + \boldsymbol{d}_H + \boldsymbol{r}_0,$$

where $\boldsymbol{d}_H$ is supported on H and

$$d_{H,t} = \boldsymbol{P}_K(W_{t-1})'(\boldsymbol{c}_1 - \boldsymbol{c}_0), \qquad t \in H.$$

Because, pointwise for $t \in H$,

$$\boldsymbol{P}_K(W_{t-1})'(\boldsymbol{c}_1 - \boldsymbol{c}_0) = m_1(W_{t-1}) - m_0(W_{t-1}) - \{r_{1,K,t} - r_{0,K,t}\},$$

Assumptions 3 and 6 imply $\|\boldsymbol{d}_H\|_\infty = O_p(1)$. Orthogonality of the candidate block sieve therefore gives

$$\boldsymbol{\mu}'\boldsymbol{M}_{K,\widetilde{\Lambda}}\boldsymbol{w} = \boldsymbol{d}_H'\boldsymbol{M}_{K,\widetilde{\Lambda}}\boldsymbol{w} + \boldsymbol{r}_0'\boldsymbol{M}_{K,\widetilde{\Lambda}}\boldsymbol{w}.$$

At the true partition,

$$\boldsymbol{\mu}'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w} = \boldsymbol{r}_0'\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{w}.$$

Subtracting and using Lemma 3,

$$\begin{aligned} T^{-1/2}|\boldsymbol{\mu}'\Delta_M\boldsymbol{w}| &\leq T^{-1/2}\|\boldsymbol{d}_H\|_1\|\boldsymbol{M}_{K,\widetilde{\Lambda}}\boldsymbol{w}\|_\infty \\ &\quad + T^{-1/2}|\boldsymbol{r}_0'\Delta_M\boldsymbol{w}| \\ &= O_p\left(\frac{b_K h}{\sqrt{T}}\right) + o_p(1). \end{aligned}$$

The case $\widetilde{k} < k_0$ is identical after exchanging the two regimes. Since $h = O_p(r_T)$, the displayed target follows.

For completeness, here is the promised sufficient verification of Assumption 10. Let $\boldsymbol{v}_k = \boldsymbol{M}_{K,k}\boldsymbol{w}$. The coefficient-update calculation in the proof of Lemma 10 shows that, for $h = |k-k_0| \leq Cr_T$,

$$\|\boldsymbol{v}_k - \boldsymbol{v}_{k_0}\|_1 = O_p\{hb_K(1+\zeta_K^2)\}.$$

Consequently,

$$T^{-1/2}|\boldsymbol{r}_0'(\boldsymbol{v}_k - \boldsymbol{v}_{k_0})| \leq T^{-1/2}\|\boldsymbol{r}_0\|_\infty\|\boldsymbol{v}_k - \boldsymbol{v}_{k_0}\|_1 = O_p\left(\frac{a_K b_K(1+\zeta_K^2)h}{\sqrt{T}}\right),$$

which is uniformly $o_p(1)$ under (3.3). $\square$

Lemma 10 (Stochastic part of partition perturbation)**.** *Under the martingale error, moment, bounded-basis, and uniform Gram assumptions, Assumption 11, and $b_K r_T/\sqrt{T} \to 0$,*

$$T^{-1/2}\boldsymbol{U}'\Delta_M \boldsymbol{w} = O_p\left(\zeta_K\sqrt{\frac{Kr_T}{T}}\right) + o_p(1).$$

In particular, this term is $o_p(1)$ if $\zeta_K\sqrt{Kr_T/T} \to 0$.

Detailed solution. It is enough to work uniformly on $|k-k_0| \le Cr_T$, because the localization event has probability tending to one for sufficiently large fixed C. Consider $k = k_0 + h > k_0$ and write

$$A = \{1,\dots,k_0\}, \quad H = \{k_0+1,\dots,k_0+h\}, \quad B = \{k_0+h+1,\dots,T\}, \quad R = H \cup B.$$

For a segment J, put

$$\boldsymbol{G}_J = \boldsymbol{P}_J'\boldsymbol{P}_J, \qquad \widehat{\boldsymbol{\gamma}}_J = \boldsymbol{G}_J^{-1}\boldsymbol{P}_J'\boldsymbol{w}_J.$$

Assumption 11 and bounded $\boldsymbol{w}$ imply $\|\widehat{\boldsymbol{\gamma}}_J\| = O_p(1)$ for every long segment. The exact update identity

$$\widehat{\boldsymbol{\gamma}}_{A\cup H} - \widehat{\boldsymbol{\gamma}}_A = \boldsymbol{G}_{A\cup H}^{-1}\boldsymbol{P}_H'(\boldsymbol{w}_H - \boldsymbol{P}_H\widehat{\boldsymbol{\gamma}}_A)$$

gives

$$\|\widehat{\boldsymbol{\gamma}}_{A\cup H} - \widehat{\boldsymbol{\gamma}}_A\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

Removing H from the true right segment gives, by the same algebra,

$$\|\widehat{\boldsymbol{\gamma}}_B - \widehat{\boldsymbol{\gamma}}_R\| = O_p\left(\frac{h\zeta_K b_K}{T}\right).$$

The true and candidate residualized weights are segmentwise

$$\boldsymbol{v}_0 = (\boldsymbol{w}_A - \boldsymbol{P}_A\widehat{\boldsymbol{\gamma}}_A, \boldsymbol{w}_R - \boldsymbol{P}_R\widehat{\boldsymbol{\gamma}}_R),$$

$$\boldsymbol{v}_k = (\boldsymbol{w}_{A\cup H} - \boldsymbol{P}_{A\cup H}\widehat{\boldsymbol{\gamma}}_{A\cup H}, \boldsymbol{w}_B - \boldsymbol{P}_B\widehat{\boldsymbol{\gamma}}_B).$$

After cancelling the $\boldsymbol{w}_H$ terms, one obtains the exact decomposition

$$\begin{aligned}\boldsymbol{U}'(\boldsymbol{v}_k - \boldsymbol{v}_0) = {} & (\boldsymbol{P}_A'\boldsymbol{U}_A)'(\widehat{\boldsymbol{\gamma}}_A - \widehat{\boldsymbol{\gamma}}_{A\cup H}) \\ & + (\boldsymbol{P}_B'\boldsymbol{U}_B)'(\widehat{\boldsymbol{\gamma}}_R - \widehat{\boldsymbol{\gamma}}_B) \\ & + (\boldsymbol{P}_H'\boldsymbol{U}_H)'(\widehat{\boldsymbol{\gamma}}_R - \widehat{\boldsymbol{\gamma}}_{A\cup H}). \end{aligned} \tag{6.1}$$

Because $\boldsymbol{P}_K(W_{t-1})U_t$ is a vector martingale difference, martingale isometry and the Gram bounds yield

$$\|\boldsymbol{P}_A'\boldsymbol{U}_A\| = O_p(\sqrt{TK}), \qquad \sup_{h\le Cr_T}\|\boldsymbol{P}_H'\boldsymbol{U}_H\| = O_p(\zeta_K\sqrt{r_T}).$$

The same bound for $\boldsymbol{P}_B'\boldsymbol{U}_B$ follows by writing it as the fixed right-segment sum minus the local H-sum. Insert these bounds into (6.1). Uniformly for $h \le Cr_T$,

$$T^{-1/2}|\boldsymbol{U}'(\boldsymbol{v}_k - \boldsymbol{v}_0)| = O_p\left(\zeta_K\sqrt{\frac{h}{T}}\right) + O_p\left(\frac{h\zeta_K b_K\sqrt{K}}{T}\right) + o_p(1).$$

The second displayed order divided by $\zeta_K\sqrt{Kh/T}$ equals $b_K\sqrt{h/T}$. Under the bridge rate $b_K r_T/\sqrt{T} \to 0$, this ratio is $o(1)$, uniformly for $h \le Cr_T$. Since $K \ge 1$, the first order is bounded by $\zeta_K\sqrt{Kh/T}$. Thus

$$T^{-1/2}\boldsymbol{U}'\Delta_M\boldsymbol{w} = O_p\left(\zeta_K\sqrt{\frac{Kr_T}{T}}\right) + o_p(1).$$

The case $k < k_0$ is obtained by reversing left and right. Notice that the proof is uniform over deterministic local candidates, so it remains valid at the data-selected $\widetilde{k}$. □

Lemma 11 (Denominator perturbation)**.** *Under the conditions of Lemmas 9 and 10, and under $\zeta_K^2 K r_T/T \to 0$,*

$$V_T(\widetilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{(1+\zeta_K)^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1) = o_p(1).$$

Detailed solution. Put $\boldsymbol{v}_k = \boldsymbol{M}_{K,k}\boldsymbol{w}$. We first reduce the feasible denominator to the oracle array $U_t v_{t,k}$, uniformly for $|k - k_0| \le Cr_T$. As in the proof of Lemma 9,

$$\boldsymbol{\mu} = \boldsymbol{Q}_{K,k}\boldsymbol{c}^{(k)} + \boldsymbol{d}_H + \boldsymbol{r}_0,$$

where $\boldsymbol{d}_H$ is supported on the boundary set and is uniformly bounded. Writing $\boldsymbol{\Pi}_k = \boldsymbol{I}_T - \boldsymbol{M}_{K,k}$,

$$\widehat{\boldsymbol{u}}(\beta_0, k) = \boldsymbol{U} - \boldsymbol{\Pi}_k\boldsymbol{U} + \boldsymbol{M}_{K,k}\boldsymbol{r}_0 + \boldsymbol{M}_{K,k}\boldsymbol{d}_H.$$

Hence

$$\widehat{\boldsymbol{Z}}(\beta_0, k) = \boldsymbol{U} \odot \boldsymbol{v}_k + \boldsymbol{\rho}_k,$$

where

$$\|\boldsymbol{\rho}_k\|_T \le \|\boldsymbol{v}_k\|_\infty \left(\|\boldsymbol{\Pi}_k\boldsymbol{U}\|_T + \|\boldsymbol{M}_{K,k}\boldsymbol{r}_0\|_T + \|\boldsymbol{M}_{K,k}\boldsymbol{d}_H\|_T\right).$$

The projection-noise proof used in Lemma 7, now uniformly over local candidates, gives $\|\boldsymbol{\Pi}_k\boldsymbol{U}\|_T = O_p(\sqrt{K/T})$. Projection contraction gives

$$\|\boldsymbol{M}_{K,k}\boldsymbol{r}_0\|_T = O_p(a_K), \qquad \|\boldsymbol{M}_{K,k}\boldsymbol{d}_H\|_T = O_p(\sqrt{h/T}).$$

Together with $\|\boldsymbol{v}_k\|_\infty = O_p(b_K)$, the sieve and bridge rates imply

$$\sup_{|k-k_0|\le Cr_T} \|\boldsymbol{\rho}_k\|_T = o_p(1). \tag{6.2}$$

Therefore it remains to compare

$$V_{U,T}(k) = T^{-1}\sum_{t=1}^{T} U_t^2 v_{t,k}^2.$$

Let $h = |k - k_0|$. The coefficient-update bounds proved in Lemma 10 imply

$$|v_{t,k} - v_{t,0}| \le \begin{cases} Cb_K, & t \in \mathcal{M}_T, \\ Ch\zeta_K^2 b_K/T, & t \notin \mathcal{M}_T. \end{cases}$$

Also $\max_t(|v_{t,k}| + |v_{t,0}|) = O_p(b_K)$. Since $T^{-1}\sum_t U_t^2 = O_p(1)$, it remains only to justify the boundary quadratic bound. Put $p = 1 + \delta_0/2 > 1$. Applied to the centered sequence $U_t^2 - \mathbb{E}(U_t^2 \mid \mathcal{F}_{t-1})$, the martingale maximal inequality on dyadic blocks gives

$$\sup_{1 \le h \le Cr_T} \frac{1}{h} \left| \sum_{t \in H_h} \{U_t^2 - \mathbb{E}(U_t^2 \mid \mathcal{F}_{t-1})\} \right| = O_p(1),$$

because the block-tail probabilities are bounded by a summable geometric series of order $2^{-j(p-1)}$. The conditional variances are bounded, so uniformly for a boundary interval H_h of length $h \le Cr_T$,

$$\sum_{t \in H_h} U_t^2 = O_p(h).$$

Consequently,

$$\begin{aligned} |V_{U,T}(k) - V_{U,T}(k_0)| &\le T^{-1} \sum_t U_t^2 |v_{t,k} - v_{t,0}|(|v_{t,k}| + |v_{t,0}|) \\ &= O_p\left(\frac{b_K^2 h}{T}\right) + O_p\left(\frac{h\zeta_K^2 b_K^2}{T}\right) + O_p\left(\frac{h^2\zeta_K^4 b_K^2}{T^2}\right). \end{aligned}$$

Assumption 11 gives $b_K^2 = O(K)$. Moreover, $h\zeta_K^2/T = o(1)$ follows from $\zeta_K^2 K r_T/T \to 0$. Consequently,

$$V_{U,T}(k) - V_{U,T}(k_0) = O_p\left(\frac{b_K^2 h}{T} + \frac{\zeta_K^2 K h}{T}\right).$$

This bound is $o_p(1)$, so $\|\boldsymbol{U} \odot \boldsymbol{v}_k\|_T = O_p(1)$ follows from the true-partition oracle LLN. Now (6.2) and the square-difference inequality give

$$V_T(k) - V_{U,T}(k) = o_p(1)$$

uniformly locally, and the same holds at k_0. Taking $h = O_p(r_T)$ proves

$$V_T(\widetilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{(1+\zeta_K)^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

□

Theorem 2 (One-break estimated-partition score equivalence)**.** *Suppose $q_0 = 1$, Assumption 7 holds with $r_T = o(\sqrt{T})$, and*

$$\zeta_K \sqrt{\frac{K r_T}{T}} \to 0, \qquad \frac{(1+\zeta_K) r_T}{\sqrt{T}} \to 0, \qquad \frac{\zeta_K^2 K r_T}{T} \to 0.$$

If Lemmas 9, 10, and 11 hold, then

$$S_T(\widetilde{\Lambda}) - S_T(\Lambda_0) = o_p(1), \qquad V_T(\widetilde{\Lambda}) - V_T(\Lambda_0) = o_p(1).$$

Detailed solution. For every partition Λ, symmetry and idempotence imply

$$S_T(\Lambda) = T^{-1/2}(\boldsymbol{U} + \boldsymbol{\mu})'\boldsymbol{M}_{K,\Lambda}\boldsymbol{w}.$$

Therefore

$$S_T(\widetilde{\Lambda}) - S_T(\Lambda_0) = T^{-1/2}\boldsymbol{\mu}'\Delta_M\boldsymbol{w} + T^{-1/2}\boldsymbol{U}'\Delta_M\boldsymbol{w}.$$

Lemma 9 bounds the first term by

$$O_p\left(\frac{b_K r_T}{\sqrt{T}}\right) + o_p(1) = o_p(1),$$

and Lemma 10 bounds the second by

$$O_p\left(\zeta_K\sqrt{\frac{K r_T}{T}}\right) + o_p(1) = o_p(1).$$

This proves the numerator equivalence.

For the denominator, Lemma 11 gives

$$V_T(\widetilde{\Lambda}) - V_T(\Lambda_0) = O_p\left(\frac{b_K^2 r_T}{T} + \frac{\zeta_K^2 K r_T}{T}\right) + o_p(1).$$

The second displayed bridge rate makes the second term $o_p(1)$. For the first term, if $r_T \geq 1$,

$$\frac{b_K^2 r_T}{T} = \frac{1}{r_T}\left(\frac{b_K r_T}{\sqrt{T}}\right)^2 = o(1).$$

Hence the denominator difference is also $o_p(1)$. □

Theorem 3 (One-break primitive estimated-partition Wilks target)**.** *Under the assumptions of Theorem 1, the convex-hull condition for the estimated-partition score array, and Theorem 2,*

$$\ell_T(\beta_0, \widetilde{\Lambda}) \Rightarrow \chi_1^2.$$

Detailed solution. Let $A_T(\Lambda) = S_T(\Lambda)$ and $B_T(\Lambda) = V_T(\Lambda)$. Theorem 2 gives

$$A_T(\widetilde{\Lambda}) = A_T(\Lambda_0) + o_p(1), \qquad B_T(\widetilde{\Lambda}) = B_T(\Lambda_0) + o_p(1).$$

The known-partition proof gives

$$A_T(\Lambda_0) \xrightarrow{\text{st}} \eta_q Z, \qquad B_T(\Lambda_0) \xrightarrow{p} \eta_q^2.$$

We must also verify the maximum condition; numerator and denominator equivalence alone would not imply it. The local representation in the proof of Lemma 11 gives

$$\widehat{\boldsymbol{Z}}(\beta_0, \widetilde{\Lambda}) = \boldsymbol{U} \odot \boldsymbol{v}_{\widetilde{k}} + \boldsymbol{\rho}_{\widetilde{k}}.$$

The envelope rate implies $\max_t |U_t v_{t,\widetilde{k}}| = o_p(\sqrt{T})$. For the remainder, the uniform projection calculation gives

$$\|\mathbf{\Pi}_{\widetilde{k}}\boldsymbol{U}\|_\infty = O_p\left(\zeta_K\sqrt{K/T}\right),$$

while the sieve calculation gives

$$\|\boldsymbol{M}_{K,\widetilde{k}}\boldsymbol{r}_0\|_\infty = O_p\{a_K(1+\zeta_K)\}.$$

Finally, because $\boldsymbol{d}_H$ has bounded entries and support of size $O_p(r_T)$,

$$\|\boldsymbol{M}_{K,\widetilde{k}}\boldsymbol{d}_H\|_\infty = O_p\left(1+\frac{r_T\zeta_K^2}{T}\right) = O_p(1).$$

Multiplication by $\|\boldsymbol{v}_{\widetilde{k}}\|_\infty = O_p(b_K)$, followed by the sieve, bridge, and envelope rates, gives $\|\boldsymbol{\rho}_{\widetilde{k}}\|_\infty = o_p(\sqrt{T})$. Hence

$$\max_t |\widehat{Z}_t(\beta_0, \widetilde{\Lambda})| = o_p(\sqrt{T}).$$

The estimated-partition convex-hull condition and Lemma 1 now yield

$$\ell_T(\beta_0, \widetilde{\Lambda}) = \frac{A_T(\widetilde{\Lambda})^2}{B_T(\widetilde{\Lambda})} + o_p(1).$$

Substitution of the two equivalences reduces this ratio to the true-partition ratio plus $o_p(1)$, and Theorem 1 gives the χ_1^2 limit. $\square$

7 Break-Date Localization

Lemma 12 (Deterministic RSS drift target)**.** *Under Assumption 12, there exists $c_\Delta > 0$ such that, uniformly over minimum-spaced one-break candidates k,*

$$\mathbb{E}\{RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0) \mid \mathcal{D}_T\} \geq c_\Delta|k-k_0|\Delta_T^2 - o_p(|k-k_0|\Delta_T^2),$$

where $\mathcal{D}_T$ denotes the design sigma-field generated by X, W and the sieve basis.

Detailed solution. The conclusion is not valid from the original martingale assumptions alone; we use Assumption 12. Let

$$\boldsymbol{y}_0 = \boldsymbol{Y} - \beta_0\boldsymbol{X}_{lag} = \boldsymbol{\mu} + \boldsymbol{U}, \qquad \boldsymbol{A}_k = \boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}.$$

Because the profile least-squares cost is the residual quadratic form,

$$RSS_K(\beta_0, k) = \boldsymbol{y}_0'\boldsymbol{M}_{K,k}\boldsymbol{y}_0.$$

Therefore

$$\Delta RSS_K(k) = \boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{\mu} + 2\boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{U} + \boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U}.$$

Conditional centering and conditional homoskedasticity give

$$\mathbb{E}\{\Delta RSS_K(k) \mid \mathcal{D}_T\} = \boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{\mu} + \sigma^2 \operatorname{tr}(\boldsymbol{A}_k).$$

Every candidate and true one-break sieve matrix has rank $2K$, so

$$\operatorname{tr}(\boldsymbol{A}_k) = \operatorname{tr}(\boldsymbol{M}_{K,k}) - \operatorname{tr}(\boldsymbol{M}_{K,k_0}) = 0.$$

The profile-identification condition (3.5) now gives

$$\mathbb{E}\{\Delta RSS_K(k) \mid \mathcal{D}_T\} \geq c_\Delta |k - k_0| \Delta_T^2 - o_p\{|k - k_0|\Delta_T^2\}.$$

To see concretely where the linear drift comes from, consider the exact-sieve case and $k = k_0 + h > k_0$. Use the sets A, H, B from the bridge proof and write $\boldsymbol{\delta} = \boldsymbol{c}_1 - \boldsymbol{c}_0$. The candidate right segment fits exactly. On the candidate left segment, after writing the common coefficient as $\boldsymbol{c}_0 + \boldsymbol{a}$, the deterministic loss is

$$L_h = \min_{\boldsymbol{a}} \{\boldsymbol{a}'\boldsymbol{G}_A\boldsymbol{a} + (\boldsymbol{a} - \boldsymbol{\delta})'\boldsymbol{G}_H(\boldsymbol{a} - \boldsymbol{\delta})\}.$$

The minimizer is

$$\boldsymbol{a}_h = (\boldsymbol{G}_A + \boldsymbol{G}_H)^{-1}\boldsymbol{G}_H\boldsymbol{\delta},$$

so

$$L_h = \boldsymbol{\delta}'\boldsymbol{G}_H\boldsymbol{\delta} - \boldsymbol{\delta}'\boldsymbol{G}_H(\boldsymbol{G}_A + \boldsymbol{G}_H)^{-1}\boldsymbol{G}_H\boldsymbol{\delta}.$$

Since $\boldsymbol{G}_A \succeq cT\boldsymbol{I}_K$ and $\lambda_{\max}(\boldsymbol{G}_H) \leq h\zeta_K^2$,

$$\boldsymbol{\delta}'\boldsymbol{G}_H(\boldsymbol{G}_A + \boldsymbol{G}_H)^{-1}\boldsymbol{G}_H\boldsymbol{\delta} \leq C\frac{h\zeta_K^2}{T}\boldsymbol{\delta}'\boldsymbol{G}_H\boldsymbol{\delta}.$$

Thus, whenever $h\zeta_K^2/T = o(1)$,

$$L_h \geq \{1 - o(1)\} \sum_{t \in H} \{\boldsymbol{P}_K(W_{t-1})'\boldsymbol{\delta}\}^2.$$

A local empirical jump-energy bound

$$h^{-1} \sum_{t \in H} \{\boldsymbol{P}_K(W_{t-1})'\boldsymbol{\delta}\}^2 \geq c\Delta_T^2$$

therefore yields the desired $ch\Delta_T^2$ drift. Approximation error and far-away candidates are precisely what the remainder $\rho_T(k)$ in Assumption 12 controls. □

Lemma 13 (RSS stochastic fluctuation target)**.** *Under Assumption 12, write*

$$\Delta RSS_K(k) = RSS_K(\beta_0, k) - RSS_K(\beta_0, k_0).$$

Uniformly over minimum-spaced one-break candidates,

$$\begin{aligned}|\Delta RSS_K(k) - \mathbb{E}\{\Delta RSS_K(k) \mid \mathcal{D}_T\}| &= O_p\Big(\sqrt{|k - k_0| K \log T}\Big) \\ &\quad + O_p(K \log T).\end{aligned}$$

Detailed solution. Continue to write $\boldsymbol{A}_k = \boldsymbol{M}_{K,k} - \boldsymbol{M}_{K,k_0}$ and $h = |k - k_0|$. From the expansion in the preceding proof,

$$\Delta RSS_K(k) - \mathbb{E}\{\Delta RSS_K(k) \mid \mathcal{D}_T\} = 2\boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{U} \tag{7.1}$$
$$+ \boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} - \mathbb{E}(\boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} \mid \mathcal{D}_T). \tag{7.2}$$

Conditional on $\mathcal{D}_T$, the first term is sub-Gaussian with variance proxy bounded by

$$C\|\boldsymbol{A}_k\boldsymbol{\mu}\|^2 \leq ChK$$

by (3.6). Hence, for a sufficiently large constant C_1,

$$\mathbb{P}\left(|2\boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{U}| > C_1\sqrt{hK\log T} \;\middle|\; \mathcal{D}_T\right) \leq 2T^{-cC_1^2}.$$

There are at most T candidate dates, so a union bound gives, uniformly in k,

$$|2\boldsymbol{\mu}'\boldsymbol{A}_k\boldsymbol{U}| = O_p\{\sqrt{hK\log T}\}.$$

For the quadratic term, $\boldsymbol{A}_k$ is symmetric and

$$\operatorname{rank}(\boldsymbol{A}_k) \leq \operatorname{rank}(\boldsymbol{\Pi}_{K,k}) + \operatorname{rank}(\boldsymbol{\Pi}_{K,k_0}) \leq 4K.$$

Also $\|\boldsymbol{A}_k\| \leq 2$ and $\|\boldsymbol{A}_k\|_F \leq 4\sqrt{K}$. The conditional Hanson–Wright inequality therefore gives, for every $x > 0$,

$$\mathbb{P}\left(|\boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} - \mathbb{E}(\boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} \mid \mathcal{D}_T)| > C\{\sqrt{Kx} + x\} \;\middle|\; \mathcal{D}_T\right) \leq 2e^{-cx}.$$

Taking $x = C_2\log T$ and applying another union bound over all dates gives

$$\sup_k |\boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} - \mathbb{E}(\boldsymbol{U}'\boldsymbol{A}_k\boldsymbol{U} \mid \mathcal{D}_T)| = O_p\{\sqrt{K\log T} + \log T\} = O_p(K\log T).$$

Combining the linear and quadratic bounds in (7.2) proves

$$|\Delta RSS_K(k) - \mathbb{E}\{\Delta RSS_K(k) \mid \mathcal{D}_T\}| = O_p\{\sqrt{hK\log T}\} + O_p(K\log T)$$

uniformly over the candidate set. This logarithmic conclusion uses the sub-Gaussian completion; it cannot in general be recovered from only the finite-moment assumption in Section 2. $\square$

Theorem 4 (One-break localization theorem target)**.** *If Lemmas 12 and 13 hold and the break signal dominates the search fluctuation, then*

$$|\widetilde{k} - k_0| = O_p(r_T),$$

where r_T is any sequence satisfying the resulting drift-fluctuation inequality. A sufficient fixed-break target is $r_T = O_p(K\log T/\Delta_T^2)$, subject to refinement after the fluctuation bound is proved.

Detailed solution. Let

$$R_T = K \log T, \qquad r_T = R_T \max\{\Delta_T^{-2}, \Delta_T^{-4}\}.$$

The sequence r_T is deterministic; the statement "$r_T = O_p(\cdot)$" in the original target should read "$r_T = O(\cdot)$." Lemmas 12 and 13 imply that, with a tight random constant $C_T = O_p(1)$,

$$\Delta RSS_K(k) \geq ch\Delta_T^2 - C_T\{\sqrt{hR_T} + R_T\} - o_p(h\Delta_T^2), \qquad h = |k - k_0|.$$

Fix $M > 0$ and suppose $h \geq Mr_T$. By the definition of r_T,

$$\frac{R_T}{h\Delta_T^2} \leq \frac{1}{M}$$

and

$$\frac{\sqrt{hR_T}}{h\Delta_T^2} = \frac{\sqrt{R_T}}{\sqrt{h}\,\Delta_T^2} \leq \frac{1}{\sqrt{M}}.$$

Choose a deterministic C so that $\mathbb{P}(C_T > C) < \varepsilon$, and then choose M so large that $C(M^{-1/2} + M^{-1}) < c/4$. On the resulting event, uniformly for $h \geq Mr_T$,

$$\Delta RSS_K(k) > 0$$

for all sufficiently large T. But the minimizer satisfies

$$\Delta RSS_K(\widetilde{k}) = RSS_K(\beta_0, \widetilde{k}) - RSS_K(\beta_0, k_0) \leq 0,$$

because k_0 is an admissible candidate. Therefore

$$\mathbb{P}\{|\widetilde{k} - k_0| > Mr_T\} \leq \varepsilon + o(1).$$

Letting first $T \to \infty$, then $M \to \infty$, proves $|\widetilde{k} - k_0| = O_p(r_T)$.

If Δ_T is bounded away from zero, then $r_T \asymp K \log T$, equivalently $r_T = O(K \log T/\Delta_T^2)$. If $\Delta_T \to 0$, the fluctuation bound as stated forces the additional Δ_T^{-4} term. The smaller rate $K \log T/\Delta_T^2$ for shrinking jumps requires a refined linear fluctuation of order $\Delta_T\sqrt{hK \log T}$. Finally, the score bridge requires the separate check

$$K \log T \max\{\Delta_T^{-2}, \Delta_T^{-4}\} = o(\sqrt{T}).$$

□

8 Unknown-Order Selection

Assumption 13 (Penalty-rate target)**.** *Let R_T denote the maximal stochastic RSS gain from searching over spurious breaks; under Lemma 13, one may take $R_T = K \log T$. The penalty satisfies*

$$\frac{\mathrm{pen}_T(q, K) - \mathrm{pen}_T(q_0, K)}{R_T} \to \infty \quad \textit{for } q > q_0,$$

and is small in absolute value relative to the deterministic RSS drift from omitting true breaks:

$$|\mathrm{pen}_T(q, K) - \mathrm{pen}_T(q_0, K)| = o(T\Delta_T^2) \quad \textit{for } q < q_0.$$

These are the rates actually used in Theorem 5.

Theorem 5 (Unknown-order consistency target)**.** *Under the localization, RSS drift, RSS fluctuation, and penalty-rate targets,*

$$\mathbb{P}\{\widehat{q}(\beta_0) = q_0\} \to 1.$$

Detailed solution. Let

$$RSS_q^* = RSS_K\{\beta_0, \widehat{\Lambda}_q(\beta_0)\}, \qquad R_T = K \log T.$$

The RSS drift and search bounds needed for order selection can be written uniformly over the fixed set $0 \le q \le \bar{q}$ as

$$\min_{q<q_0}(RSS_q^* - RSS_{q_0}^*) \ge cT\Delta_T^2 - O_p(R_T), \tag{8.1}$$

$$\max_{q>q_0}(RSS_{q_0}^* - RSS_q^*) = O_p(R_T). \tag{8.2}$$

The first display says that at least one true jump is omitted; the resulting deterministic loss is of order $T\Delta_T^2$. The second says that an extra, spurious split can exploit noise only at the searched stochastic order.

The penalty condition in the original workbook must be sharpened as follows: for every $q > q_0$,

$$\frac{\mathrm{pen}_T(q, K) - \mathrm{pen}_T(q_0, K)}{R_T} \longrightarrow \infty,$$

and for every $q < q_0$,

$$|\mathrm{pen}_T(q, K) - \mathrm{pen}_T(q_0, K)| = o(T\Delta_T^2).$$

Mere divergence of the overfitting penalty is not enough when $R_T \to \infty$.

For $q < q_0$, subtract the true-order penalized criterion:

$$\begin{aligned} &RSS_q^* + \mathrm{pen}_T(q, K) - RSS_{q_0}^* - \mathrm{pen}_T(q_0, K) \\ &\quad \ge cT\Delta_T^2 - O_p(R_T) - o(T\Delta_T^2) > 0 \end{aligned}$$

with probability tending to one, because $R_T = o(T\Delta_T^2)$. For $q > q_0$,

$$\begin{aligned} &RSS_q^* + \mathrm{pen}_T(q, K) - RSS_{q_0}^* - \mathrm{pen}_T(q_0, K) \\ &\quad \ge -O_p(R_T) + \{\mathrm{pen}_T(q, K) - \mathrm{pen}_T(q_0, K)\} > 0 \end{aligned}$$

with probability tending to one. Since $\bar{q}$ is fixed, a finite union bound proves

$$\mathbb{P}\{\widehat{q}(\beta_0) = q_0\} \to 1.$$

A transparent sufficient penalty is

$$\mathrm{pen}_T(q, K) = qb_T, \qquad K \log T \ll b_T \ll T\Delta_T^2.$$

Such a sequence exists because break detectability gives $K \log T = o(T\Delta_T^2)$; for example, $b_T = \sqrt{(K \log T)(T\Delta_T^2)}$ satisfies the two rate inequalities. □

Corollary 1 (Unknown-order Wilks). *Suppose Theorem 5 and the fixed-order estimated-partition Wilks theorem hold. Then*

$$\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\} \Rightarrow \chi_1^2.$$

Detailed solution. Let

$$E_T = \{\widehat{q}(\beta_0) = q_0\}.$$

By Theorem 5, $\mathbb{P}(E_T) \to 1$. On E_T, the definitions and the common deterministic tie-breaking rule give

$$\widehat{\Lambda}(\beta_0) = \widehat{\Lambda}_{q_0}(\beta_0) = \widetilde{\Lambda}.$$

For every continuity point x of the χ_1^2 distribution,

$$\begin{aligned}&\left|\mathbb{P}\left[\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\} \le x\right] - \mathbb{P}\left[\ell_T(\beta_0, \widetilde{\Lambda}) \le x\right]\right| \\ &\quad \le \mathbb{P}(E_T^c) \to 0.\end{aligned}$$

The fixed-order estimated-partition Wilks theorem then yields the claimed χ_1^2 limit. □

9 Local Alternatives

Assumption 14 (Local-law oracle and drift). *Let $\beta_T = \beta_0 + d_T$, $d_T \to 0$, and define*

$$g_{t,T} = d_T(\boldsymbol{M}_{K,\Lambda_0}\boldsymbol{X}_{lag})_t v_{t,0}.$$

Under the local law P_{β_T},

$$T^{-1/2}\sum_{t=1}^{T} U_t v_{t,0} \xrightarrow{\text{st}} \eta_q Z, \qquad T^{-1}\sum_{t=1}^{T} U_t^2 v_{t,0}^2 \xrightarrow{p} \eta_q^2, \qquad \max_t |U_t v_{t,0}| = o_p(\sqrt{T}),$$

and

$$T^{-1/2}\sum_{t=1}^{T} g_{t,T} \xrightarrow{p} \eta_q \delta_h, \qquad T^{-1}\sum_{t=1}^{T} g_{t,T}^2 = o_p(1), \qquad \max_t |g_{t,T}| = o_p(\sqrt{T}).$$

Lemma 14 (Local shifted-score replacement target). *Suppose Assumptions 6, 3, 5, and 9 continue to hold under P_{β_T}. Then*

$$T^{-1/2}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})\} = o_p(1), \tag{9.1}$$

$$T^{-1}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0, \Lambda_0)^2 - (U_t v_{t,0} + g_{t,T})^2\} = o_p(1), \tag{9.2}$$

$$\max_t |\widehat{Z}_t(\beta_0, \Lambda_0) - (U_t v_{t,0} + g_{t,T})| = o_p(\sqrt{T}). \tag{9.3}$$

Detailed solution. Under P_{β_T},

$$\boldsymbol{Y} - \beta_0 \boldsymbol{X}_{lag} = \boldsymbol{Q}_{K,\Lambda_0}\boldsymbol{c}_0 + \boldsymbol{r}_0 + \boldsymbol{U} + d_T \boldsymbol{X}_{lag}.$$

Apply $\boldsymbol{M}_{K,\Lambda_0}$ and put

$$\boldsymbol{e}_0 = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{r}_0, \qquad \boldsymbol{h}_0 = \boldsymbol{\Pi}_{K,\Lambda_0}\boldsymbol{U}, \qquad \boldsymbol{x}_0^c = \boldsymbol{M}_{K,\Lambda_0}\boldsymbol{X}_{lag}.$$

Then

$$\widehat{\boldsymbol{u}}(\beta_0, \Lambda_0) = \boldsymbol{U} - \boldsymbol{h}_0 + \boldsymbol{e}_0 + d_T \boldsymbol{x}_0^c.$$

Multiplying componentwise by $\boldsymbol{v}_0$ gives the exact identity

$$\widehat{Z}_t(\beta_0, \Lambda_0) = U_t v_{t,0} + g_{t,T} + (e_{t,0} - h_{t,0})v_{t,0}.$$

The final term is exactly the same feasibility error as under the null. The proof of Lemma 7 therefore gives

$$\|(\boldsymbol{e}_0 - \boldsymbol{h}_0) \odot \boldsymbol{v}_0\|_T = o_p(1), \qquad \|(\boldsymbol{e}_0 - \boldsymbol{h}_0) \odot \boldsymbol{v}_0\|_\infty = o_p(\sqrt{T}),$$

and the numerator projection cancellation plus the projection-bias condition gives its normalized sum as $o_p(1)$. This proves the first and third displays of the lemma.

For the square replacement, put $a_{t,T} = U_t v_{t,0} + g_{t,T}$ and $d_{t,T} = (e_{t,0} - h_{t,0})v_{t,0}$. Then

$$T^{-1}\sum_t \{\widehat{Z}_t^2 - a_{t,T}^2\} = 2T^{-1}\sum_t a_{t,T}d_{t,T} + T^{-1}\sum_t d_{t,T}^2.$$

Assumption 14 gives

$$\|\boldsymbol{U} \odot \boldsymbol{v}_0\|_T = O_p(1), \qquad \|\boldsymbol{g}_T\|_T = o_p(1),$$

so $\|\boldsymbol{a}_T\|_T = O_p(1)$. Cauchy–Schwarz and $\|\boldsymbol{d}_T\|_T = o_p(1)$ make both terms on the right $o_p(1)$, proving the second display. □

Theorem 6 (Local noncentral chi-square target)**.** *Under Assumption 14, Lemma 14, the local convex-hull condition, and the estimated-partition bridge under P_{β_T},*

$$\ell_T(\beta_0, \widetilde{\Lambda}) \Rightarrow \chi_1^2(\delta_h^2).$$

If $\mathbb{P}\{\widehat{q}(\beta_0) = q_0\} \to 1$, the same limit holds for $\ell_T\{\beta_0, \widehat{\Lambda}(\beta_0)\}$.

Detailed solution. First work at the true partition. Lemma 14 and Assumption 14 give

$$\begin{aligned} T^{-1/2}\sum_t \widehat{Z}_t(\beta_0, \Lambda_0) &= T^{-1/2}\sum_t U_t v_{t,0} + T^{-1/2}\sum_t g_{t,T} + o_p(1) \\ &\xrightarrow{\text{st}} \eta_q(Z + \delta_h). \end{aligned}$$

For the second moment,

$$\begin{aligned} T^{-1}\sum_t (U_t v_{t,0} + g_{t,T})^2 &= T^{-1}\sum_t U_t^2 v_{t,0}^2 \\ &\quad + 2T^{-1}\sum_t U_t v_{t,0} g_{t,T} + T^{-1}\sum_t g_{t,T}^2. \end{aligned}$$

The first term converges to η_q^2, and the last is $o_p(1)$. By Cauchy–Schwarz,

$$\left| T^{-1} \sum_t U_t v_{t,0} g_{t,T} \right| \le \left(T^{-1} \sum_t U_t^2 v_{t,0}^2 \right)^{1/2} \left(T^{-1} \sum_t g_{t,T}^2 \right)^{1/2} = o_p(1).$$

The local replacement lemma therefore yields

$$T^{-1} \sum_t \widehat{Z}_t(\beta_0, \Lambda_0)^2 \xrightarrow{p} \eta_q^2.$$

Moreover,

$$\max_t |U_t v_{t,0} + g_{t,T}| = o_p(\sqrt{T}),$$

and the maximum replacement error is also $o_p(\sqrt{T})$. The local convex-hull condition and Lemma 1 imply

$$\ell_T(\beta_0, \Lambda_0) \Rightarrow \frac{\eta_q^2 (Z + \delta_h)^2}{\eta_q^2} = (Z + \delta_h)^2 \sim \chi_1^2(\delta_h^2).$$

Under the local-law version of the one-break bridge, the estimated-partition numerator and denominator differ from their true-partition counterparts by $o_p(1)$, and the estimated score maximum is $o_p(\sqrt{T})$. Applying the same scalar EL expansion transfers the limit to $\ell_T(\beta_0, \widetilde{\Lambda})$. Finally, on the event $\{\widehat{q}(\beta_0) = q_0\}$, the unknown-order and fixed-order partitions coincide. If that event has probability tending to one under P_{β_T}, the same event-transfer argument as in Corollary 1 proves the final assertion. □

10 Master Dependency Checklist

Goal	Must prove first	Output when finished
Known-partition Wilks	Lemmas 4, 5, 6, 7	Theorem 1
One-break bridge	Lemmas 9, 10, 11	Theorem 2
Break localization	Lemmas 12, 13	Theorem 4
Unknown order	Theorem 5 and penalty-rate target	Corollary 1
Local alternatives	Local oracle, shifted replacement, local bridge	Theorem 6

Recommended homework order

1. Prove deterministic projection algebra and leverage bounds.
2. Prove known-partition Wilks from primitive martingale assumptions.
3. Prove the one-break partition perturbation bounds.
4. Prove profile-RSS break localization.
5. Only then work on unknown-order selection and local alternatives.